

SIYI LIU

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Department of Psychology, Virginia Polytechnic Institute and State University
890 Drillfield Drive, Blacksburg, VA 24060

EDUCATION

Doctor of Philosophy, Industrial/Organizational Psychology, Virginia Tech Expected 2028 Spring
Thesis (Defended): The Performance of Multimodal Large Language Models in Visual Personnel Assessments
Advisor : Dr. Louis Hickman

Master of Finance, University of Maryland, College Park (UMD) 2022 - 2023
(A dual M.S. program between UMD and SUFE)

Master of Finance, Shanghai University of Finance and Economics (SUFE) 2020 - 2022
Winner of *National Graduate Scholarship in China* (Selected in the Top **3%**)
Thesis: Analyzing Risk Factors for Bond Default in Real Estate Industry Using a Random Forest Model

Bachelor of Mathematical Finance, Wuhan University of Technology 2016 - 2020

SKILLS

Research Methods and Statistics: Regression analysis, ANOVA, Meta-analysis, Factor analysis, Structural equation modeling, Mediation and moderation analysis, Natural language processing, Machine learning

Softwares: R (advanced), Python (advanced), SPSS (advanced), STATA (intermediate), SQL (intermediate), Power BI (novice), Tableau (novice); Bloomberg (intermediate); Adobe Premiere (intermediate)

Financial Modelling: Asset Pricing, Corporate Valuation, Comparable Company Analysis, Equity Analysis

Languages: English, Chinese (native), Spanish (basic/DELE A1 Certificate)

RESEARCH INTERESTS

1. Implementation of Machine Learning (ML), Natural Language Processing (NLP), and Large Language Models (LLMs) in Organizations (e.g., Personnel Selection, Assessment and Feedback)
2. Psychometrics and Research Methodology
3. Mitigating Adverse Impact in Personnel Selection and Assessment
4. Application of ML, NLP, and LLMs in Clinical Interventions

ACCEPTED PUBLICATIONS

Focus: Research Methodology

1. **Liu, S.**, Hickman, L., Dahlander, L., & Piezunka, H. (2026). Textual similarity in organizational research: Review of applications, consistency of methods, and best practice recommendations. *Organizational Research Methods*. <https://doi.org/10.1177/10944281261432629>

Focus: Application of LLMs in Personnel Assessment

1. Stockdale, K.S., Hickman, L., & **Liu, S.** (2026). Scoring employment interviews with large language models: Components of structure, validity investigations, and best practice recommendations. *Journal of Applied Psychology*. <https://doi.org/10.1037/apl0001396>

Focus: Application of ML, NLP, and LLMs in Clinical Interventions

1. Hudson, C., Hickman, L., **Liu, S.**, Young, H., Sabbagh, M., & Harkness, K. (2026). Assessing individual differences in adult theory of mind through mental state language: Human and artificial intelligence approaches. *Assessment*. [Under journal production]

Focus: Diversity–Validity Trade-Off in Selection

1. Hickman, L., Speer, A., Song, C., & **Liu, S.** (2025). Exploring how multi-objective optimization reduces Black-White group differences in Artificial Intelligence-based essay scores. [Received first round of review & resubmission in *Journal of Applied Psychology*].

Focus: Application of LLMs in Personnel Assessment

1. **Liu, S.**, Hickman, L., & Liff, J. (2026). The performance of multimodal large language models in visual personnel assessments. [Received first round of review & resubmission in *International Journal of Selection and Assessment*].
2. Bakeman, T., Shaban T., Hickman, L., **Liu, S.**, & Rose, M. (2026). The performance of multimodal large language models in visual personnel assessments. [Under review in *Journal of Applied Psychology*].

Focus: Application of ML, NLP, and LLMs in Clinical Interventions

1. Penedo, J. M. G., Coyne, A., **Liu, S.**, Ronchi, A., Manubens, R., Areas, M., & Hickman, L. (2026). Using natural language processing and large language models to scale-up coding of psychotherapist interventions in Latin America. [Received first round of review & resubmission for Special Issue on Artificial Intelligence in Psychotherapy Research in *Journal of Consulting and Clinical Psychology*].

SELECTED MANUSCRIPTS IN PROGRESS

Focus: Application of AI in Organizational Feedback

1. **Liu, S.** (2026). From performance to well-being: an integrative conceptual review of AI-powered feedback in organizations. [Target: *Journal of Applied Psychology*]

BOOK CHAPTERS

1. Hickman, L., Stockdale, K.S., & **Liu, S.** (2026). Large Language Models in Recruitment and Selection. *Research Handbook on Recruitment and Selection*. [Under production]

CONFERENCE PRESENTATIONS AND POSTERS

Under Review

Accepted

1. **Liu, S.** (2026, September). From Performance to Well-Being: An Integrative Conceptual Framework of AI-Powered Feedback in Organizations. 2026 SIOP People Analytics LEC Conference, Baltimore, MD, United States.
2. Hickman, L., Speer, A., Song, C., & **Liu, S.** (2026, August). Exploring how multi-objective optimization reduces group differences in automatic essay scores. 86th Annual Meeting of the Academy of Management, Philadelphia, PA, United States.
3. **Liu, S.**, Hickman, L., & Liff, J. (2026, May). The performance of multimodal large language models on visual personnel assessments: Initial evidence and implications of cheating with AI. In Campion, E. D. (Chair), Misbehavin': Implications of AI Use in Unproctored Pre-Hire Assessments [Symposium]. Society for Industrial and Organizational Psychology Annual Conference, New Orleans, LA, United States.
4. **Liu, S.** (2026, February). The performance of multimodal large language models in visual personnel assessments. Poster presented at 2026 Graduate Research Symposium at William & Mary, Williamsburg, VA, United States.
5. **Liu, S.**, Hickman, L., Dahlander, L., & Piezunka, H. (2025, April). Textual similarity in organizational research: Review of applications, consistency of methods, and best practice recommendations. In L. Hickman (Co-Chair) M. Liu (Co-Chair), Machine Learning for I-O 7.0: Natural Language Processing in Research and Assessment [Symposium]. Society for Industrial and Organizational Psychology Annual Conference, Denver, CO, United States.

6. **Liu, S.**, Hickman, L., Dahlander, L., & Piezunka, H. (2025, April). Textual similarity in organizational research: Review of applications, consistency of methods, and best practice recommendations. Paper presented at the 41st Annual GPSS Research Symposium and Exposition: Future of Research: Empowerment, Innovation, and Education, Virginia Tech, Blacksburg, VA, United States.
7. Stockdale, K.S., **Liu, S.**, Ansari, A., & Hickman, L. (2024, August). Scoring interviews with large language models: Initial validity evidence. Applications of NLP to Human Resource Management: From Word Dictionaries to Large Language Models [Symposium]. 84th Annual Meeting of the Academy of Management, Chicago, IL, United States.
8. Penedo, J. M. G., Coyne, A., **Liu, S.**, Ronchi, A., Manubens, R., Areas, M., & Hickman, L. (2024, June). Natural language processing to identify cognitive-behavioural vs. psychodynamic interventions: Automatizing micro-analytic process evaluations in psychotherapy. Paper presented at the Society for Psychotherapy Research 55th Annual International Meeting, Ottawa, Canada.
9. Stockdale, K. S., **Liu, S.**, Ansari, A., & Hickman, L. (2024, April). Scoring interviews with large language models: Initial validity evidence. In L. Hickman M. Liu (Chairs), Machine learning for I-O 6.0: Applications of large language models [Symposium]. Society for Industrial and Organizational Psychology Annual Conference, Chicago, IL, United States.

RESEARCH EXPERIENCE

Graduate Student Researcher in Workplace Assessment and Social Perception Lab Aug 2023 - Present
Virginia Tech, Department of Psychology *Blacksburg, VA*

- Led and executed two first-author research projects and contributed to five additional collaborative studies.
- Developed AI-driven personnel selection and assessment tools (ML, NLP, and LLMs), with emphasis on explainability, bias mitigation, and real-world deployment.
- Prepared conference presentation materials for SIOP submissions and coordinated reviewer responses during review and resubmission processes.
- Collaborated with an external talent assessment vendor on a master's thesis examining the implications of AI use in multimodal personnel testing.
- Conducted analyses and co-authored a successful \$20,000 APA Psychotherapy Research Grant proposal.

Graduate Student Researcher Sept 2020 - May 2022
Shanghai University of Finance and Economics, School of Finance *Shanghai, China*

- Collected, preprocessed, and feature-engineered large-scale macroeconomic, corporate financial and bond-level datasets for a master's thesis.
- Developed and tuned a Random Forest classification model in thesis to predict real estate bond default risk, achieving 96.2% accuracy and 0.92 AUC; evaluated model performance using ROC analysis and implemented model interpretability techniques including Gini importance and SHAP values for feature attribution and risk factor analysis.
- Conducted an empirical quantitative finance study on Asset-Backed Notes (ABN) pricing by designing a stochastic cash-flow forecasting framework; implemented the Cox–Ingersoll–Ross (CIR) interest rate model, Monte Carlo simulation, and Option-Adjusted Spread (OAS) methodology to simulate interest-rate paths, optimize pricing, and assess model sensitivity and risk exposure.
- Built a hybrid decision-support and performance evaluation model for a Chinese bailout fund by integrating multi-source economic, financial, and survey data; applied expert-weighted feature aggregation using the Intuitionistic Fuzzy Analytic Hierarchy Process (IFAHP) to construct interpretable composite indicators; then co-authored a 40,000-word data-driven research report supporting policy and investment decision-making.

PROFESSIONAL EXPERIENCE

Graduate Consulting Intern Aug 2024 - Present
Virginia Tech, Department of Organizational Excellence *Blacksburg, VA*

- Collaborated with a consulting team of five to develop Virginia Tech's Organizational Excellence Model, including subscale item design; built an interactive Power BI dashboard for visualization and stakeholder use.

- Generated evidence-based policy recommendations for university leadership, derived from cross-university benchmarking and data analyses.
- Synthesized and delivered external research insights aligned with the university's strategic priorities and context.
- Coordinated with consulting team members and cross-departmental stakeholders to scope, plan, and implement university strategic projects.

Economic Affairs Intern

Aug 2022 - Sept 2022

United Nations Economic and Social Commission for Asia and the Pacific, Transport Division

Remote

- Conducted comparative literature reviews assessing sustainability progress across 12 Northeast and South Asian countries, synthesizing cross-national evidence.
- Performed qualitative policy analysis on electric mobility and sustainable transport across ESCAP member states, identifying key themes and policy patterns.
- Authored an 11,000-word analytical report on financing mechanisms and market development in Shanghai's electric mobility sector.

Overseas Investment Intern

July 2022 - Aug 2022

Contemporary Amperex Technology Co. Limited (CATL), Corporate Planning Department

Ningde, China

- Analyzed macroeconomic and industry trends affecting CATL's lithium-ion battery portfolio, synthesizing insights to support strategic decision-making.
- Maintained and updated monthly lithium market databases tracking the commercial development of 50+ global lithium mining projects, ensuring data accuracy and consistency.
- Conducted due diligence research on potential lithium acquisition targets using structured analytical frameworks (e.g., SWOT and economic analysis).
- Supported internal valuation and decision processes by compiling, validating, and organizing financial data for DCF analyses in collaboration with senior team members.

Bond Issuance Intern

Jul 2021 - Oct 2021

Haitong Securities, Debt Capital Markets Department

Shanghai, China

- Prepared and revised application and approval documents for three bond projects in Shandong Province (total value over RMB 5 billion), including bond prospectuses and advocacy reports, to meet rigorous document analysis and regulatory compliance.
- Conducted due diligence research on a target company in Renxing, China by reviewing financial and organizational information, reassessing bond eligibility, and coordinating with internal teams and external clients.
- Drafted and reviewed issuance and announcement materials, including pricing analyses; supported project tracking by updating two semi-annual corporate bond reports.
- Supported project development and team collaboration by contributing to financing proposals and pitchbooks for new bond initiatives, including preliminary issuance plans and customized solutions for issuers.

Equity Analyst Intern

Feb 2021 - May 2021

CITIC Securities, Transportation Group of Research Department

Shanghai, China

- Collected, tracked, and analyzed industry data on a daily basis to monitor key indicators, and maintained internal express, freight, and shipping databases.
- Conducted industry and policy research in the transportation sector; authored weekly and monthly research reports and analytical reviews of annual reports for listed express companies.
- Contributed to the preparation of annual report summary presentations by analyzing companies' business models, operational and financial performance; produced daily trend reports for clients.
- Participated in private interviews and strategy meetings with major industry companies to gather up-to-date insights, synthesized findings; communicated key trends to the team leader, supporting team decision-making.

AWARDS AND HONORS

SIOP Conference Student Travel Award (\$500)

2026

Society for Industrial and Organizational Psychology

New Orleans, LA

National Graduate Scholarship in China (\$3000) Shanghai University of Finance and Economics, School of Finance	2021 <i>Shanghai, China</i>
First-class Graduate Scholarship (\$1700) Shanghai University of Finance and Economics, School of Finance	2021 <i>Shanghai, China</i>
Second-class Graduate Scholarship (\$1100) Shanghai University of Finance and Economics, School of Finance	2020 <i>Shanghai, China</i>
Excellence Award (Selected in the Top 10%) 6th National Master of Finance Case Study Competition in China	2020 <i>China</i>
First-class Undergraduate Scholarship (\$420) Wuhan University of Technology, School of Economics	2019 <i>Wuhan, China</i>
Second-class Undergraduate Scholarship (\$300) Wuhan University of Technology, School of Economics	2017 - 2018 <i>Wuhan, China</i>

PROFESSIONAL, INSTITUTIONAL & COMMUNITY SERVICE

Peer Reviewer <i>Psychotherapy Research</i>	2026
Peer Reviewer <i>Organizational Research Methods</i>	2026
Peer Reviewer <i>Academy of Management</i> , 2026 Annual Meeting	2026
Ad Hoc Reviewer <i>Journal of Applied Psychology</i> , with Dr. Louis Hickman	2025
Panelist at Research in STEM Fields Tech Girls U.S. Program, Virginia Tech	2025
Graduate Student Committee Member Department of Psychology, Virginia Tech	2025

PROFESSIONAL AFFILIATIONS

Society for Industrial and Organizational Psychology, *Member*
Academy of Management, *Member*

PROFESSIONAL REFERENCES

Dr. Louis Hickman
Assistant Professor, Industrial-Organizational Psychology, Department of Psychology, Virginia Tech
Email: louishickman@vt.edu

Dr. Ivan Hernandez
Associate Professor, Industrial-Organizational Psychology, Department of Psychology, Virginia Tech
Faculty Affiliate, Center for the Future of Work Places and Practices, Virginia Tech
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Dr. Ning Hsu
Assistant Professor, Industrial-Organizational Psychology, Department of Psychology, Virginia Tech
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